

Erika and Tim wrote the expression $\frac{(-2x^{-4} \cdot x^2)^3}{4x^{-5}}$ in different ways and arrived at different expressions. Their work is shown.

Erika's Work	Tim's Work
$\frac{(-2x^{-4} \cdot x^2)^3}{4x^{-5}}$ $\frac{-8x^{-12} \cdot x^6}{4x^{-5}}$ $\frac{-8x^{-72}}{4x^{-5}}$ $\frac{-2}{x^{67}}$	$\frac{(-2x^{-4} \cdot x^2)^3}{4x^{-5}}$ $\frac{(-2x^{-2})^3}{4x^{-5}}$ $\frac{-8x^{-6}}{4x^{-5}}$ $\frac{-2}{x}$

1. Even though one student's resulting expression is not equivalent to $\frac{(-2x^{-4} \cdot x^2)^3}{4x^{-5}}$, identify which laws of exponents each student applied in what order.

Order of Laws Applied	Erika	Tim
First		
Second		
Third		

2. Which student made an error?
3. Write a note to this student explaining their mistake and how to correct their error.

For each expression in questions 4–10, write an equivalent expression using positive exponents and the fewest factors as possible.

4. $\frac{8w^{-6}p}{-2w^{-4}p^{-5}}$

5. $\frac{k^{-1}j^{-6}}{7k^3j^{-2}}$

6. $\left(\frac{-5f^{-6}g^3}{2f^7g^{-3}}\right)^2$

7. $(m^0 \cdot m^{-9})^{-4}$

8. $c^{-5} \cdot \frac{3c^{-9}}{-6c^5}$

9. $\left(\frac{3m^8n^{-9}}{-8m^{-11}n^2}\right)^0$

10. $\frac{-7d^{-5} \cdot d^4}{11d^{-19}}$